

2025 State Tournament - Div C -Georgia

Entomology - EXAM



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School Name/Team: _____

Student Names: _____

Instructions:

1. Write all answers on the Answer Sheet. DO NOT WRITE on Test Packet.
2. See the Image Sheet when asked to refer to an image by a question (it may be a separate sheet or attached to the back of this packet).
3. For numerical questions, answers within a reasonable range will be accepted.
4. When answering free-response questions, use the appropriate abbreviations for satellite and instrument names
5. For free-response questions, the answer sheet indicates the amount of space you should allocate to fully answer each question. Points are assigned based on accurately answering each part of the question.
6. Ties marked with a * will be broken in order

FOR GRADERS ONLY: _____ / 33 (Total points)

1. A fossil insect has halteres on the hindwings but membranous forewings with venation. What order is it most likely? (1 point)
 - a. Mecoptera
 - b. Diptera
 - c. Trichoptera
 - d. Hymenoptera
2. An insect with dense coeloconic sensilla on its antennae is most specialized in detecting: (1 point)
 - a. Colony pheromones
 - b. Humidity and temperature gradients
 - c. Plant alkaloids
 - d. Low-frequency vibrations
3. If the galea and lacinia fuse into a piercing stylet, this feeding style most resembles which order? (1 point)
 - a. Coleoptera
 - b. Hemiptera
 - c. Lepidoptera
 - d. Diptera
4. Two insects have 100 Hz wingbeats. One has synchronous flight muscles, the other asynchronous. Which requires more neural impulses per second? (1 point)
 - a. Synchronous — resonance is absent
 - b. Asynchronous — indirect muscles fire continuously
 - c. Asynchronous — impulses must match each wingbeat
 - d. Synchronous — every beat needs an impulse
5. An insect has aquatic larvae with tracheal gills and adults with hairy wings held rooflike. Which order? (1 point)
 - a. Lepidoptera
 - b. Trichoptera
 - c. Neuroptera
 - d. Ephemeroptera
6. A South American beetle closely resembles an Australian beetle; DNA suggests divergence 100 mya. What explains this? (1 point)
 - a. Parallel evolution
 - b. Convergent evolution
 - c. Gondwanan distribution via continental drift
 - d. Human introduction

7. Imaginal discs are best described as: (1 point)
- a. Developmental compartments absent in ametabolous insects
 - b. Delayed metamorphic tissues
 - c. Regeneration centers in ametabolous insects
 - d. Retained larval features into adulthood
8. A flower with deep tubular corollas, strong nocturnal odor, and night nectar is most adapted for: (1 point)
- a. Bees
 - b. Beetles
 - c. Moths
 - d. Flies
9. A beetle larva is discovered under bark. It has a hard head capsule, chewing mouthparts, and three pairs of thoracic legs, but no prolegs. What type of larva is this? (1 point)
- a. Campodeiform
 - b. Eruciform
 - c. Scarabaeiform
 - d. Vermiform
10. Which of the following insects is a keystone pollinator in desert ecosystems? (1 point)
- a. Yucca moths (Prodoxidae)
 - b. Carpenter bees (Xylocopinae)
 - c. Army ants (Ecitoninae)
 - d. Cicada killers (Sphecius speciosus)
11. Which is TRUE regarding ametabolous development? (1 point)
- a. Insects with this type of development include mayflies and dragonflies.
 - b. Juveniles resemble adults but lack wings and genitalia.
 - c. The order Zygentoma is an example.
 - d. It is the most common type of insect development.
12. Which insect group is the primary vector of Chagas disease? (1 point)
- a. Blackflies (Simuliidae)
 - b. Kissing bugs (Reduviidae: Triatominae)
 - c. Sand flies (Psychodidae: Phlebotominae)
 - d. Mosquitoes (Culicidae)

Match the following families with their correct order: (1 point each)

13. Formicidae _____

A. Coleoptera

14. Tabanidae _____

B. Hymenoptera

15. Cicadellidae _____

C. Diptera

16. Meloidae _____

D. Hemiptera

17. What order do dragonflies belong to? (1 point)

- a. Isoptera
- b. Mallophaga
- c. Hemiptera
- d. Odonata

18. What is the collection method for adult flying insects? (1 point)

- a. Cast net
- b. Drift net
- c. Fine net
- d. Hand net

19. Apterygota is best defined as: (1 point)

- a. Wings develop inside of body in immature insects and aren't visible until the adult immatures from the pupa
- b. Adult like immature without wings
- c. The wings develop externally on the nymph body
- d. Adults with wings

20. Exopterygota is best defined as: (1 point)

- a. Wings develop inside the body in immature insects and aren't visible until the adult immatures from the pupa
- b. Adult like immature without wings
- c. The wings develop externally on the nymph body
- d. Adults with wings

21. How many pairs of legs are attached to the thorax? (1 point)

- a. Zero
- b. One
- c. Two
- d. Three

22. What disease is related to fleas? (1 point)

- a. Plague
- b. Malaria
- c. Typhus
- d. Lyme disease

23. What disease is related to mosquitos? (1 point)

- a. Plague
- b. Dengue fever
- c. Lyme disease
- d. Typhus

24. What disease is related to ticks? (1 point)

- a. Plague
- b. Malaria
- c. West Nile virus
- d. Lyme disease

Questions 25-27 refer to the image below.



25. What is this insect's order and common name? (2 points)

26. What kind of habitat is the nymph of this insect mostly found in? Give an example. (2 points)

27. Name two common predators for the adult version of this insect. (1 point)

28. Match each insect mouth to its function by its letter beside its corresponding function. (4 points; 1 point each)

a.



b.



c.



d.



I . Sucking _____

II . Sponging _____

III . Piercing-Sucking _____

IV . Syphoning _____